

TORRINGTON, WY, USA

WMO: 725763

Lat: 42.061N

Lon: 104.158W

Elev: 4199

StdP: 12.60

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 94053

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.7	1.0	-11.0	3.5	-4.0	-5.4	4.8	4.1	33.2	39.6	29.0	37.6	5.2	300	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.5	97.0	63.3	93.5	63.0	90.6	62.7	68.5	85.2	67.0	84.2	65.6	82.4	10.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	103.8	73.1	62.3	98.3	72.1	60.6	92.4	71.1	35.1	84.8	33.8	84.0	32.7	82.3	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.5	25.1	21.6	DB	-17.2	101.9	6.1	2.3	-21.6	103.6	-25.2	104.9	-28.6	106.2	-33.1	107.8	(4)
(5)			WB	-17.2	72.1	5.7	1.4	-21.3	73.1	-24.6	73.9	-27.8	74.7	-32.0	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	29.2	29.4	39.6	47.0	55.7	67.7	74.8	71.7	62.8	48.7	37.6	28.0		
	DBStd	18.82	11.41	12.18	10.71	8.82	8.72	6.86	5.20	5.50	7.80	9.21	10.40	12.18		
	HDD50	2970	645	580	343	154	46	0	0	0	6	131	383	682		
	HDD65	6414	1109	998	786	541	306	49	4	13	131	508	823	1146		
	CDD50	2774	1	2	22	64	222	530	770	673	389	90	10	1		
	CDD65	742	0	0	0	1	16	129	309	222	64	1	0	0		
	CDH74	10430	0	0	21	85	436	1919	3842	2782	1196	147	2	0		
	CDH80	4993	0	0	1	11	137	877	2076	1356	504	31	0	0		
(14) Wind	WSAvg	9.5	9.9	10.1	10.5	11.3	10.2	9.4	8.4	8.2	8.2	8.9	9.0	9.5		
(15) Precipitation	PrecAvg	13.4	0.3	0.4	0.7	1.6	2.4	2.2	1.8	1.1	1.2	0.9	0.5	0.3		
	PrecMax	20.7	0.9	1.3	2.8	3.5	6.1	5.3	4.4	2.7	4.7	3.3	1.3	1.0		
	PrecMin	7.0	0.0	0.0	0.1	0.2	0.3	0.1	0.0	0.0	0.2	0.0	0.0	0.0		
	PrecStd	2.9	0.2	0.3	0.6	0.8	1.6	1.4	1.1	0.8	1.0	0.7	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	65.9	77.2	81.5	89.2	98.9	101.9	98.1	94.4	83.9	72.1	60.1		
		MCWB	43.1	45.7	50.4	52.2	58.9	62.7	64.7	62.2	62.1	55.2	48.7	43.0		
	2%	DB	55.1	59.1	70.9	75.9	83.9	93.5	98.4	94.7	90.4	78.4	65.9	55.1		
		MCWB	40.3	42.4	47.7	51.0	58.1	61.9	64.1	62.6	59.6	53.1	46.3	40.7		
	5%	DB	51.2	53.7	64.4	71.6	79.1	89.6	95.1	91.5	86.2	72.8	61.2	50.7		
		MCWB	38.4	39.6	45.3	49.9	56.9	61.8	63.8	62.9	58.8	51.4	44.4	38.5		
	10%	DB	46.4	48.0	59.3	65.7	73.4	84.9	91.2	88.2	81.4	66.8	55.5	46.4		
		MCWB	36.2	36.8	43.2	48.0	54.9	60.8	64.1	62.7	58.1	49.4	41.5	36.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	46.2	51.2	55.1	64.0	69.3	71.0	69.8	66.2	57.4	50.2	44.0		
		MCDB	58.3	64.0	73.1	74.5	81.5	84.4	88.2	85.6	85.8	76.5	68.9	58.5		
	2%	WB	41.2	42.6	48.5	52.3	61.0	66.5	69.0	67.7	63.4	55.1	47.2	41.4		
		MCDB	53.8	57.3	67.6	71.2	76.1	82.5	85.8	84.2	81.3	73.1	64.7	54.1		
	5%	WB	38.9	40.2	46.0	50.5	58.7	64.4	67.7	66.1	61.6	52.8	44.8	39.0		
		MCDB	49.7	52.8	63.0	68.2	73.2	81.2	85.5	82.5	78.6	69.2	59.9	49.9		
	10%	WB	36.6	37.0	43.7	48.6	56.5	62.7	66.3	64.5	59.9	50.4	42.3	36.5		
		MCDB	46.5	47.1	58.2	64.2	71.0	79.8	83.5	80.7	75.4	65.2	55.0	46.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.3	24.4	27.1	26.8	26.4	29.7	30.5	30.6	30.9	28.8	26.9	23.8		
		MCDBR	29.5	31.5	37.4	38.5	35.8	36.7	36.7	35.8	38.8	40.1	35.6	29.2		
		MCWBR	17.4	17.7	18.5	17.3	14.3	12.2	10.2	10.6	13.5	18.6	19.0	17.5		
	5% WB	MCDBR	27.9	30.5	34.7	34.5	31.2	32.5	30.4	30.2	32.8	35.7	34.3	27.5		
		MCWBR	17.1	17.6	17.7	16.4	13.8	12.8	10.7	10.8	12.3	17.1	18.7	17.1		
(40) Clear-Sky Solar Irradiance	taub		0.245	0.255	0.277	0.305	0.325	0.335	0.360	0.354	0.317	0.278	0.259	0.247		
	taud		2.588	2.554	2.521	2.452	2.463	2.457	2.422	2.424	2.511	2.590	2.558	2.559		
	Ebn at Noon		297	309	311	306	301	297	288	287	292	295	287	286		
	Edh at Noon		21	25	29	34	35	35	36	35	30	24	21	19		
(44) All-Sky Solar Radiation	RadAvg		699	977	1370	1678	1935	2297	2263	1975	1597	1095	766	584		
	RadStd		30	60	101	104	161	129	88	60	93	93	43	33		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page